

Worksheet: Material balances on an absorption column

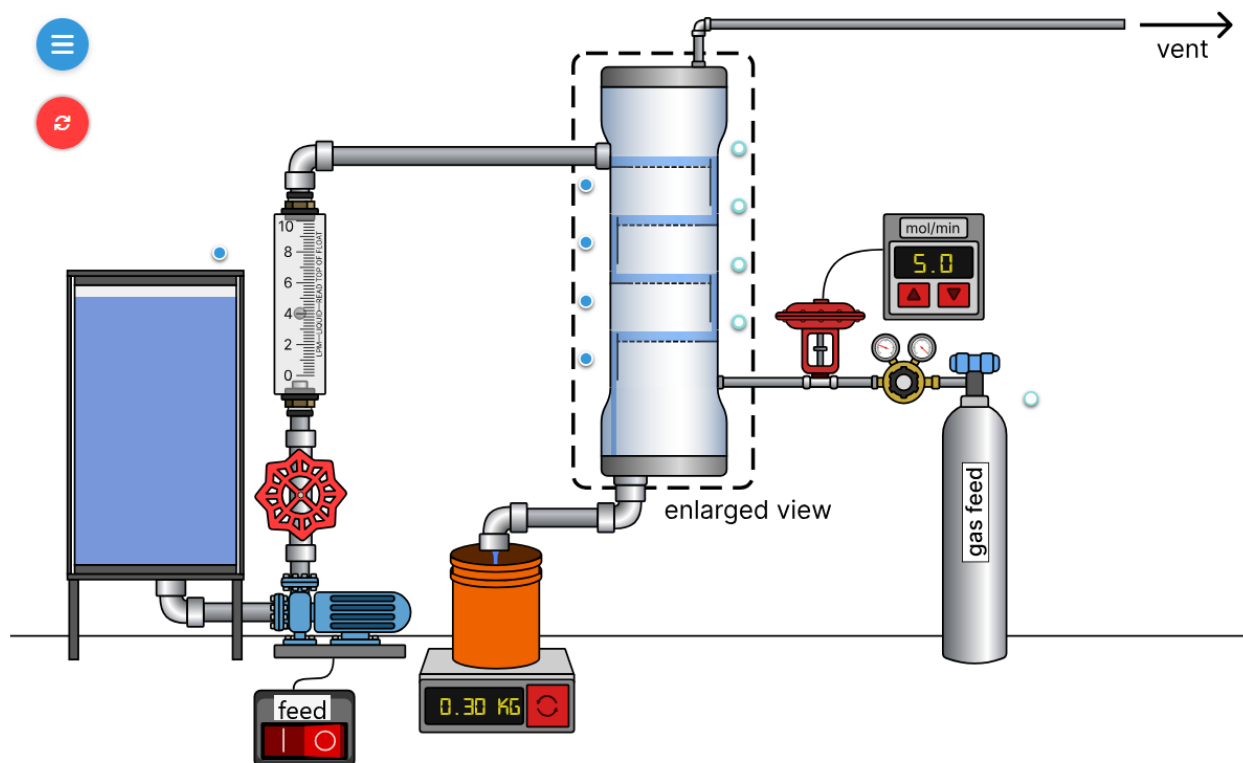
Name(s) _____

This experiment applies material balances to a trayed absorption column. The absorption column removes an impurity solute from a gas stream by absorbing the impurity into a liquid stream. The column is a steady-state, counter-flow experiment at constant pressure. The exit streams from each stage are in equilibrium.

Student learning objectives

1. Be able to apply material balances to a system.
2. Be able to explain what happens in a absorption column.
3. Be able to apply the relation between gas-phase mole ratio and liquid-phase mole ratio (Henry's law).

Equipment



The system consists of stages where the liquid phase (water) flows down, and the gas phase flows up. The phases leaving each stage are in phase equilibrium. The top stage is stage 1 and stage numbers increase moving down the column. The equilibrium is described by

$$\text{Henry's law: } y_i P = H x_i$$

where y_i is the mole ratio of solute in the gas leaving stage i and x_i is the mole ratio of solute in the liquid phase leaving stage i .

H = Henry's constant (bar) P = pressure (bar)

Because the impurity solute concentrations are low, the liquid and gas flow rates are approximately constant from stage to stage.

Questions to answer before starting the experiment

1. How many independent material balances can you write for the absorption column? Explain.
2. The two phases contact each other in a counter-current flow. What might be the advantage of counter-current flow over co-current flow?
3. Where is the highest concentration of solute in the absorption column, at the top or the bottom of the column? Why?

Experimental Procedure:

1. Use the menu on the top left to select the absorption column mode and the number of stages.
2. Switch on the liquid feed pump (left) to fill the column. Then, drag and rotate the red feed valve to change the liquid flow rate. Record the inlet volumetric flow rate, then calculate the mass flow rate, and enter the values in Table 1.
3. Click on the blue cylinder valve to open it and pressurize the line. Rotate the pressure regulator valve to adjust the pressure. Record the pressure at the top of Table 3. Use the buttons on the flow controller to adjust the gas flow rate and record the value in Table 2.
4. Allow the system to reach steady state. View liquid and vapor solute mole ratios by hovering over the blue dots (liquid) and white dots (gas) next to the column. Steady state can be assumed if the solute mole ratios change by less than 2% in 30 s.
5. Using the weight scale at the liquid outlet, record the starting weight in Table 1 and start a timer. After some time has elapsed, record the weight and time in Table 1. Calculate the liquid mass flow rate out and record it in Table 1. Calculate the difference between inlet and outlet liquid flow rates in Table 1.
6. Enter the liquid flow rate in g/min in Table 2 and calculate the flow rate in mol/min; the liquid is water.

Table 1 Liquid flow rate

Volumetric flow rate in (L/min)	Mass flow rate in (kg/min)	Starting weight (kg)	Stopping weight (kg)	Elapsed time (s)	Mass flow rate out (kg/min)	Mass flow rate difference (in – out) kg/min

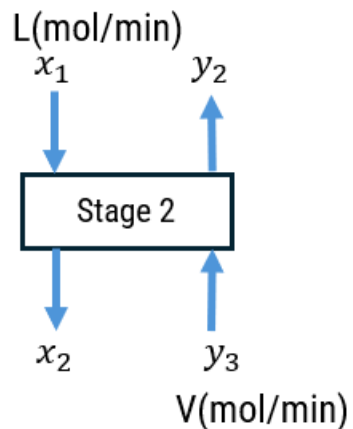
Table 2 Liquid and gas flow rates

L(g/min)	L(mol/min)	V(mol/min)

7. Move the mouse over the buttons above the liquid and vapor feeds and record their solute mole ratios in Table 3. Likewise, record the solute mole ratios at the exit of each stage in Table 3. The figure for stage 2 shows what the subscripts refer to in the stage.

Table 3 Solute mole ratios Pressure _____ bar

Stage	Solute mole ratio in liquid (ppm)	Solute mole ratio in gas (ppm)
Liquid feed	$x_0 =$	
1	$x_1 =$	$y_1 =$
2	$x_2 =$	$y_2 =$
3	$x_3 =$	$y_3 =$
4	$x_4 =$	$y_4 =$
5	$x_5 =$	$y_5 =$
Gas feed		$y_6 =$



where L = liquid molar flow rate
 V = gas molar flow rate

Data Analysis:

Carry out mole balances on the solute for each stage to determine if mole balances are satisfied. Table 4 shows the molar flow rates used for stage 2. Enter the mole balance values in Table 5.

Table 4 Notation for calculating mole balances

	Solute in ($\mu\text{mol}/\text{min}$)			Solute out ($\mu\text{mol}/\text{min}$)			Difference	
Stage	Liquid	Gas	Total	Liquid	Gas	Total	(In – out)	% difference
2	x_1L	y_3V	$x_1L + y_3V$	x_2L	y_2V	$x_2L + y_2V$	$(x_1L + y_3V) - (x_2L + y_2V)$	

Table 5 Solute mole balance

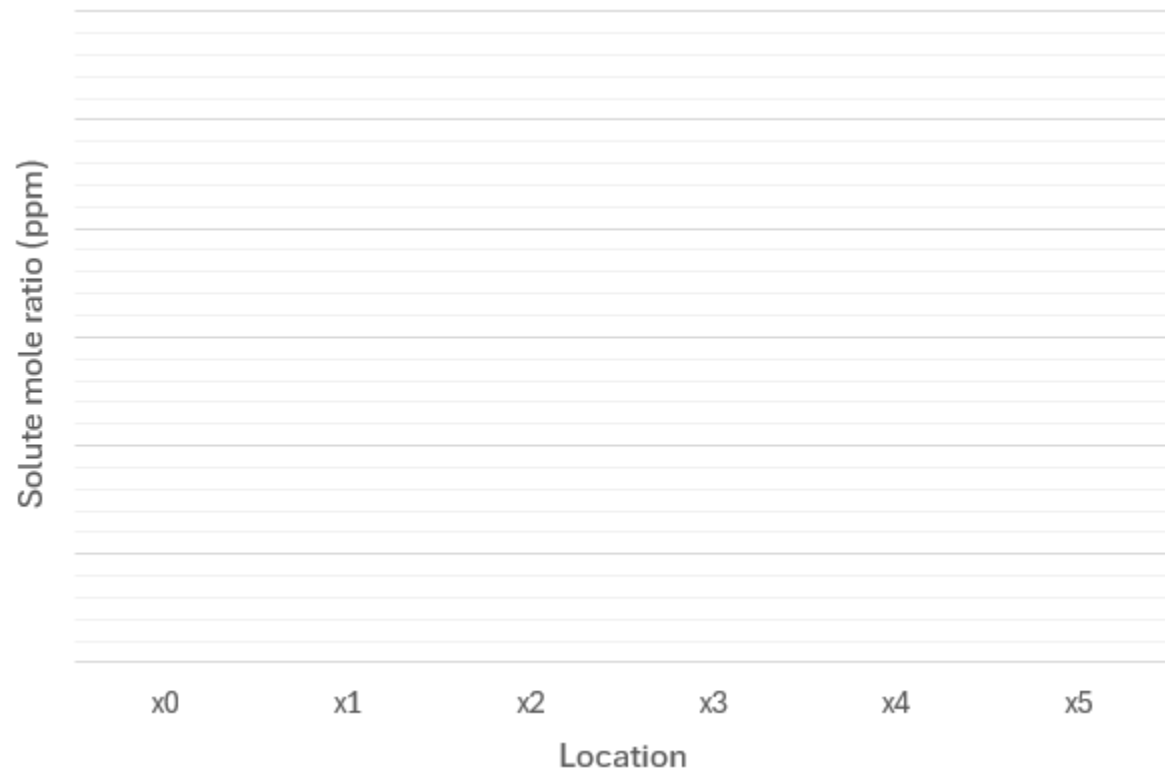
	Solute in ($\mu\text{mol}/\text{min}$)			Solute out ($\mu\text{mol}/\text{min}$)			Difference	
Stage	Liquid	Gas	Total	Liquid	Gas	Total	(In – out)	% difference
1								
2								
3								
4								
5								

The streams leaving each stage (e.g., x_2 and y_2 for stage 2) are in equilibrium. Determine Henry's law constant for each stage from the equation $y_iP = Hx_i$ and enter the values in Table 3.

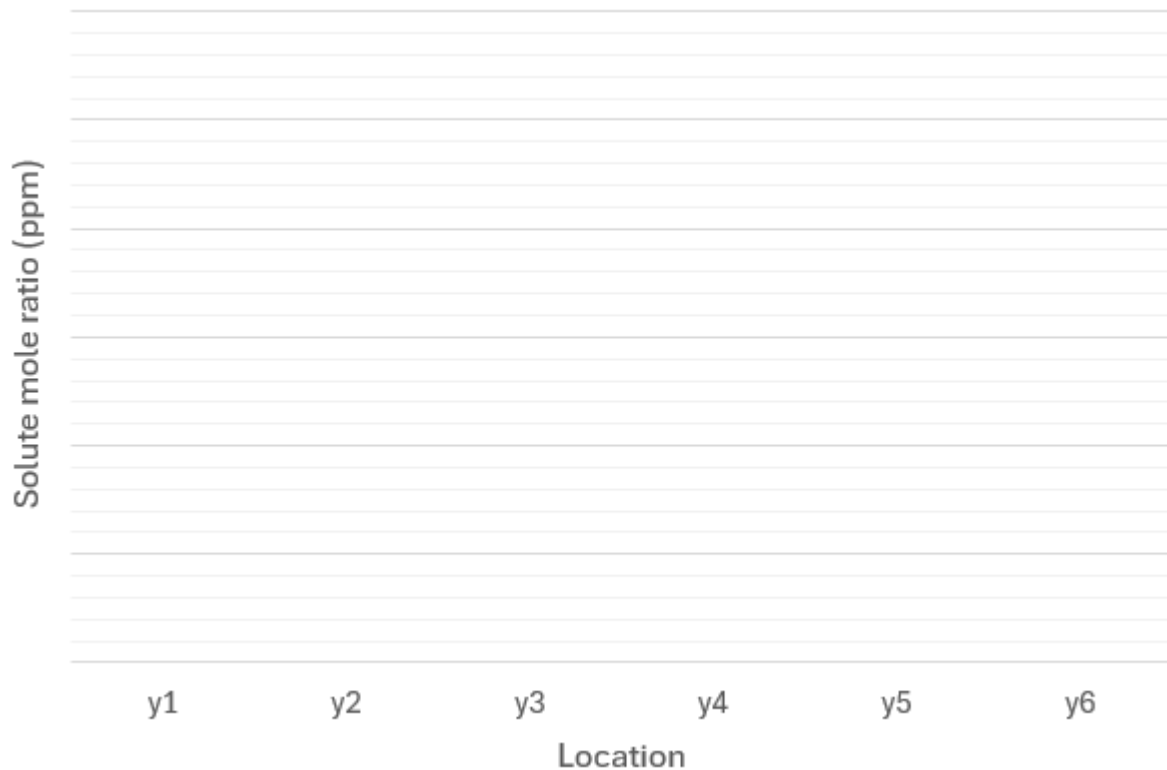
Table 6 Henry's law constant

Stage	H (bar)
1	
2	
3	
4	
5	
average	

Plot the solute mole ratios for the gas phase and connect the points with lines to clearly show the trends. The x_0 notation is the liquid feed mole ratio into stage 1.



Plot the solute mole ratios for the liquid phase and connect the points with lines to clearly show the trends. The y_{n+1} notation, where n is the number of stages, is the gas feed mole ratio into stage n.



Questions to answer

1. Where might these measurements have errors?
2. What safety precautions would you take to conduct this experiment in the laboratory?
3. To make the process more effective, would you run at low pressure or high pressure? Why? Change the pressure in the experiment and observe the effect on mole ratios.
4. To make this process more effective, would you run at low temperature or high temperature? Why?